

ShowMeAround: Giving Virtual Tours Using Live 360 Video

Alaeddin Nassani

The University of Auckland
alaeddin.nassani@auckland.ac.nz

Huidong Bai

The University of Auckland
huidong.bai@auckland.ac.nz

Li Zhang

The University of Auckland
zhlnpu@gmail.com

Mark Billingham

The University of Auckland
mark.billinghurst@auckland.ac.nz

ABSTRACT

This demonstration presents ShowMeAround, a video conferencing system designed to allow people to give virtual tours over live 360-video. Using ShowMeAround a host presenter walks through a real space and can live stream a 360-video view to a small group of remote viewers. The ShowMeAround interface has features such as remote pointing and viewpoint awareness to support natural collaboration between the viewers and host presenter. The system also enables sharing of pre-recorded high resolution 360 video and still images to further enhance the virtual tour experience.

CCS CONCEPTS

• **Human-centered computing** → Collaborative and social computing; Collaborative and social computing systems and tools.

KEYWORDS

360-degree video, Virtual Tours, Augmented Reality

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1 INTRODUCTION

This demonstration introduces ShowMeAround, a video conferencing platform designed for giving live 360 video tours to small groups of people. In 2020, in response to the global COVID-19 pandemic, most international travel was significantly reduced, and tourism businesses began to explore online options for remote tours. For example, AirBnB introduced AirBnB Online Experiences [1] for small group tours, while Amazon Explore [2] supports one-on-one virtual experiences. In both cases the experiences are delivered over a desktop video conferencing platform, but one of the limitations of this is that participants can only get a limited view of the presenter's surroundings, and so don't really feel that they are really in the remote location.

This lack of immersion and situational awareness is a recognized problem with traditional video conferencing and so researchers have explored a number of ways to address this limitation. One approach is to use live streamed 360-degree video, which enables the remote viewer to have an independent view into the sender's location. For example, JackInHead [10] captured videos from multiple head worn cameras and stitched them together into a live 360-degree video that a remote user could be immersed in through a Virtual Reality (VR) head mounted display (HMD). Similar live streaming 360-degree video interfaces have been developed by number of other researchers [4–8]. Commercial systems such as Periscope [3], Facebook Live [11] and Avatour [9] also provide support for live 360-degree video streaming. However, these systems are typically designed for delayed video broadcast to many viewers (Periscope, Facebook) or real time streaming to a limited number of people (less than five for Avatour). In both cases there also is relatively limited opportunity for feedback, either through only using emoticons and text (Periscope, Facebook live), or audio chat (Avatour).

In our research we are exploring technology for supporting small group (10-30 people) remote experiences, such as guided tours in tourism. In this paper we report on ShowMeAround, a platform for supporting 360 video conferencing for small groups. Unlike previous research and commercial applications, ShowMeAround has a number of novel features, including (1) it provides a 360 video plug-in for a well-established and easy-to-use video conferencing platform, (2) it scales to support live 360 video streaming to 20 or more viewers through only a web browser, (3) the remote viewers are able to provide virtual pointing feedback, (4) the viewpoints of the remote viewers are shown, (5) pre-recorded content can be combined with the live 360 video, and (5) the ability to overlay Augmented Reality (AR) virtual cues on the live 360 video. Overall, the main difference between this and other systems is that ShowMeAround has been specifically designed for small group 360-video tours and so includes features to enhance the tour experience.

In this paper we describe the prototype platform developed and the various components and features it has. We then discuss the user experience of people trying the demonstration, and close with a brief conclusion and directions for future work.

2 THE SHOWMEAROUND PROTOTYPE

In terms of hardware, the ShowMeAround prototype consists of a Ricoh Theta V 360 camera [16] connected to a laptop computer with audio headphones and a microphone. This is carried by the host presenter as they walk through the space. The 360-video from

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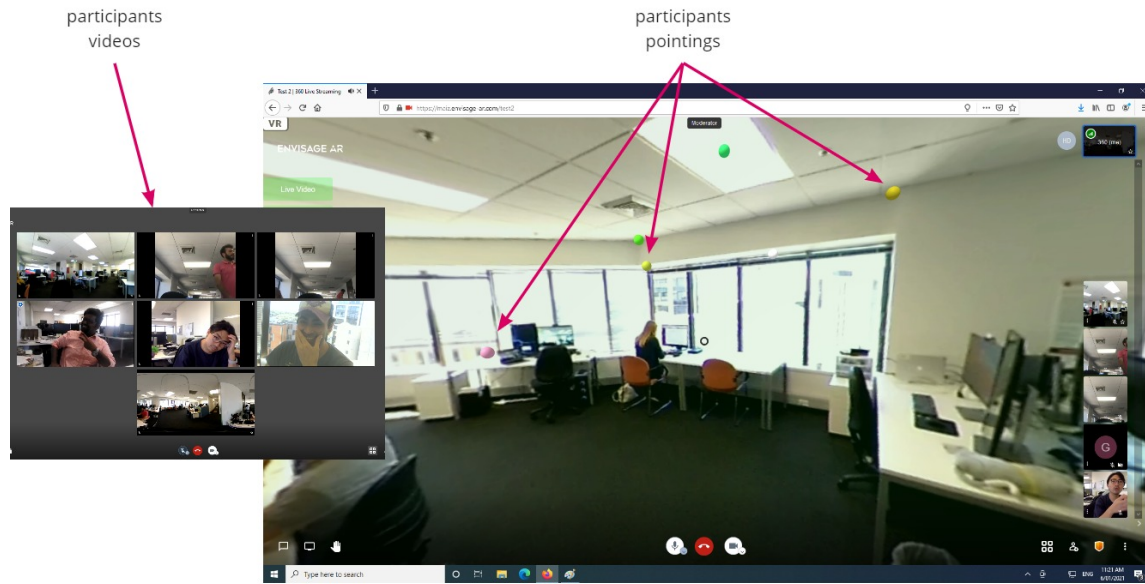


Figure 1: The ShowMeAround user interface: (left) conference view of all participants. (right) 360 view showing dots with different colours indicating where participants are pointing.

the Theta V is wirelessly streamed to the ShowMeAround software running on a server and then to all of the viewers connected through a web interface. On the laptop, the host presenter can also connect to the server and see video from all of the connected viewers and talk to them.

The ShowMeAround software is based on the Jitsi open-source video conferencing platform [12, 14] which enables multiple audio/video/text connections as well as peer-to-peer connections. Jitsi supports multi-platform implementations including web, mobile and desktop with back-end hosted on a Linux-based operating system. It enables users to join a multi-party video conference purely through accessing a web URL, with no application download needed. The Jitsi platform consists of a number of core components; 1) nginx, a web https server, 2) Jitsi-meet [15], a front-end WebRTC application that handles delivering video-stream connections to the web using real time protocol, 3) prosody, an XMPP server for connection signaling and establishing connections, and 4) jicofo, a conference management tool and videobridge for the WebRTC backend for routing and managing video streams.

Our system extends the web-based implementation of the Jitsi platform by rendering and sharing a 360-degree video stream from the presenter as a Jitsi-meet plug-in. This was done by writing react-native code and using the Jitsi-meet API. Using our plug-in, the viewers can share their pointing and point-of-view to facilitate conversation with the presenter. In addition to the live 360-degree video stream, users can choose to view static (pre-recorded) 360-degree videos and images. The A-Frame WebGL framework can also be used as a layer to render 3D objects on the video layer [13] to provide an Augmented Reality overlay on the viewers view. We

have implemented a number of other features, as detailed below. Figure 1 shows the ShowMeAround interface.

Rendering 360-degree video stream: The video stream is rendered as a material for a spherical 3D object which allows viewers to navigate and pan around different sides of the 360-degree video stream. The 360-degree video rendering is enabled for the host of the conference. All other participants in the conference (viewers) are streaming their video in standard 2D video streams. The viewers can pan around the 360-video by clicking and dragging with their mouse. In this way they can easily see the view into the remote space that interests them and have a different viewpoint orientation than the host.

Video resolution: To enhance the resolution of the 360-degree video, the video quality of viewers' video stream is limited to standard definition (SD, ~360p) while the host presenter's video remains unlimited to high-definition (HD, ~1080p) based on the camera resolution.

Pointing: The viewers can click on any part of the 360-degree video image to indicate points of interest during the conversation with the host. The pointing position is marked by a virtual dot rendered in 3D space. The position of the dot is calculated based on the intersection between mouse click event and an invisible 3D sphere which enables the dot to remain attached to the background of the 360-degree video stream. The position of the dot in 3D space is sent to all viewers using WebRTC data channel (in x, y, z) which allows all of the viewers and the host to see a different coloured dot for each conference user (viewer/host) (see Figure 2).

Point of view: Similar to pointing, the user's point of view (i.e., which part of the 360-degree video is being viewed) is also shared between users using the WebRTC data channel (in x, y, z). The point

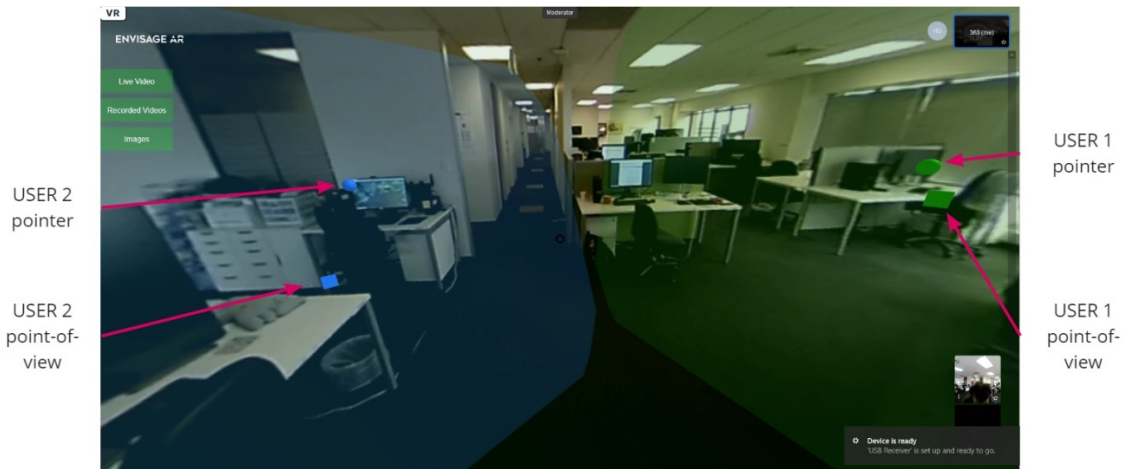


Figure 2: Sharing point of view of the remote viewers. Semi-transparent rectangles indicate the point of view of remote viewers. In this example here the green rectangle is one user's viewpoint, while the blue is another user.

of view of the remote users is represented by a semi-transparent rectangle rendered on top of the 360-degree video that moves around when the corresponding remote viewer pans around. The colour of the rectangle matches the colour of the pointing dot (see Figure 2).

User interface: The ShowMeAround application (Figure 1) shows the participants in a grid-like conference meeting. Users can choose to maximise the view of any of the video streams including the 360-degree video stream. On the 360-video, the user can use a mouse click to leave a pointing dot. They also can choose from the left-hand menu to view pre-recorded 360 static content (images/videos). Pre-recorded 360 video and still images can be loaded and used in the host presentation. This has the advantage of allowing the viewers to see higher resolution still imagery of points of interest, or video of activity that is currently not happening in the presenter space.

Host Control: Although viewers can normally freely pan around the 360-video, there may be times when the host presenter wants everyone to look in the same direction. This can be done by the host taking control of all of the viewer's viewpoints and dragging them around with their mouse motion. When control is released the viewers can return to having independent control.

Multi-Device Support: ShowMeAround is a WebGL application, and so can be viewed on any device with a web browser. The interface is designed for desktop use, but it can also be viewed on a mobile phone, tablet or VR HMD. For example, if viewed in the Oculus Quest HMD the user will be immersed in a 360-video view and the head tracking of the Quest can be used to set their viewpoint, allowed the user to naturally turn in the real world to see different parts of the live 360-video.

3 THE USER EXPERIENCE

Since ShowMeAround is a video conferencing experience it can be demonstrated to viewers from anywhere in the world through their own desktop browser. Viewers will be sent a web url and when they load the link in their web browser, they will immediately be taken to the ShowMeAround desktop interface (See Figure 1). They

will see video streams from all of the other participants in the tour and will be able to talk to them like they normally would in a video conferencing application (e.g. Zoom). However, one of the video windows will be a 360-video from the host presenter. When the viewer double clicks on this it will expand to fill their whole screen and they will be able to see a live 360-video stream which they can pan around and view from any direction using their mouse.

When they see an object of interest, the viewer can point on it using mouse input. They will also be able to see the pointing actions of other viewers on the tour, and a representation of the viewpoints of each of the people on the tour (see figure 2). At different points in the tour the host will load pre-recorded 360-video and high-resolution imagery. This will enable the viewers to see objects of interest at higher resolution than what is possible using the live 360-video. They will also be able to talk to the host throughout the tour and ask questions about what they are seeing.

There will be two people giving the tour (See figure 3). One person will be holding the laptop with Ricoh Theta V and microphone/speakers attached. The second will be appearing before the camera and giving the tour, enabling them to speak freely without being encumbered by any equipment.

For the purpose of the demonstration, a 10-15 minute tour will be given through our laboratory. The overall impression for the viewer is that they can feel like they are on a real tour, especially because they can look around independently of the host presenter and explore the space. The 360-video will provide a high degree of immersion, even on a desktop computer, although those participants with access to a VR HMD could have an even more immersive experience. At the end of the tour, we will have a question-and-answer session to answer any of the technical or application questions coming from participants.

4 CONCLUSION

This work introduces ShowMeAround a 360-degree virtual tour experience based on the open-source Jitsi video conferencing platform. The system enables a small group of viewers to experience

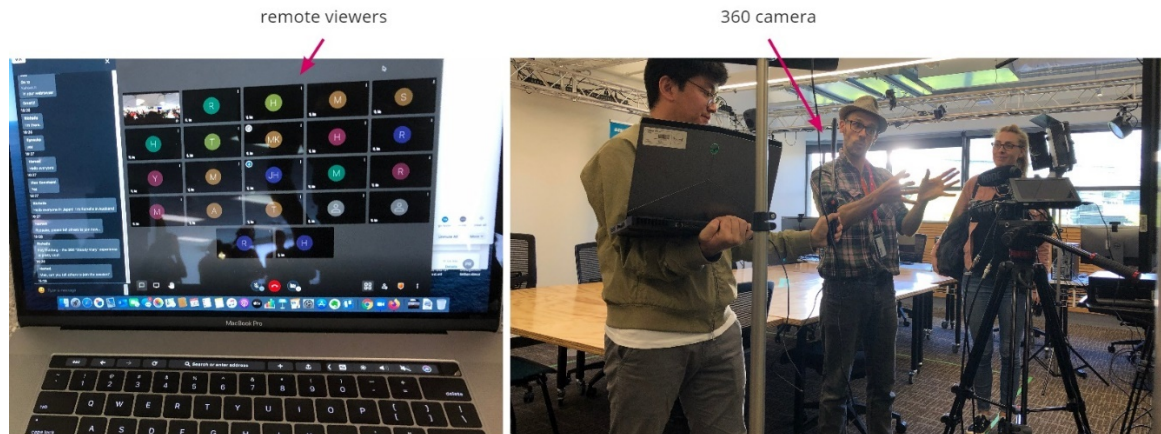


Figure 3: The ShowMeAround remote user interface (left) and the local guide (right)

a virtual tour together with a live presenter, looking around the remote space, sharing their point of view and pointing on top of the shared 360 video stream to facilitate better communication with the presenter.

Although ShowMeAround is designed to support virtual tours and tourism, it could be used in a much wider range of applications where remote users want to share a space together. For example, for education and training, business meetings, site inspections, watching performances, and other domains.

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